FPR versus SSP comparison for screening statistics

mfscreen2 simulation

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Contents

1 Competing Screening Statistics

The simulation compares three marginal screening procedures. Each procedure assigns one score to each predictor variable, then selects variables according to a threshold indexed by a nominal level q.

1.1 Reciprocal model-free screening statistic

For predictor X_j, let

$$z_{ij} = X_{ij} - \bar{X}_j, \qquad u_i = Y_i - \bar{Y}.$$

The `mfscreen` statistic is

$$D_j^2 = \frac{\sum_i z_{ij}^2 u_i^2}{\left(\sum_i z_{ij} u_i\right)^2} - \frac{2\sum_i z_{ij}^3 u_i}{\left(\sum_i z_{ij} u_i\right)\left(\sum_i z_{ij}^2\right)} + \frac{\sum_i z_{ij}^4}{\left(\sum_i z_{ij}^2\right)^2}.$$

It is the squared reciprocal of the robust marginal screening statistic:

$$D_j^2 = \frac{1}{T_j^2}.$$

Small values indicate stronger marginal evidence. For each nominal level q, we set

$$\gamma_q = \Phi^{-1}(1 - q/2)$$

and select variable j when

$$D_j^2 \le \frac{1}{\gamma_q^2}.$$

1.2 Pearson SIS

Pearson SIS is the classical marginal correlation screen. For each predictor, we compute the Pearson correlation r_j between X_j and Y, then use the usual two-sided correlation test statistic

$$t_j = |r_j|\sqrt{\frac{n-2}{1-r_j^2}}.$$

The two-sided p-value is

$$p_j = 2P(t_{n-2} \ge t_j).$$

For nominal level q, Pearson SIS selects predictor j when

$$p_j \le q.$$

1.3 Spearman SIS

Spearman SIS applies the same p-value rule after replacing each variable by its ranks. It is a rank-based marginal screen and is less sensitive to monotone transformations than Pearson correlation. For nominal level q, it selects predictor j when the two-sided rank-correlation p-value satisfies

$$p_j^{\text{rank}} \le q.$$

1.4 How the plotted points are constructed

For a fixed method and a fixed nominal level q, each Monte Carlo replication produces a selected set $\widehat{S}_q$. Let S be the true active set and S^c its complement. In replication b,

$$\mathrm{FPR}_b(q) = \frac{|\widehat{S}_q^{(b)} \cap S^c|}{|S^c|}$$

and

$$\mathrm{SSP}_b(q) = 1\{S \subseteq \widehat{S}_q^{(b)}\}.$$

The point plotted in the FPR–SSP plane is

$$\left(\frac{1}{B}\sum_{b=1}^{B}\mathrm{FPR}_b(q), \frac{1}{B}\sum_{b=1}^{B}\mathrm{SSP}_b(q)\right),$$

where B is the number of Monte Carlo replications. The labels on selected points show the corresponding nominal q level.

2 Simulation Setup

The simulation uses fixed sample size

$$n = 2000$$

and number of candidate predictors

$$p = 1000.$$

The true active set is

$$S = \{1, 2, 3, 4, 5\}.$$

The number of Monte Carlo replications is

$$B = 100.$$

The q grid is

$$0.001, 0.005, 0.01, 0.025, 0.05, 0.1, 0.2, 0.3.$$

Predictors are generated independently as

$$X_{ij} \sim N(0, 1).$$

2.1 Parametric linear model

The parametric model is

$$Y = 0.22X_1 - 0.20X_2 + 0.18X_3 - 0.16X_4 + 0.14X_5 + \varepsilon,$$

with

$$\varepsilon \sim N(0, 2.3^2).$$

This is a deliberately weak linear signal, so the curves are not saturated at SSP = 1.

2.2 Nonparametric nonlinear model

The nonparametric model is

$$Y = 0.35\{1.6 \sin(X_1) + 1.2 \tanh(X_2) + 0.9X_3^3/(1 + X_3^2) - 0.8 \exp(0.35X_4) + 0.7X_5\} + \sigma(X)\varepsilon,$$

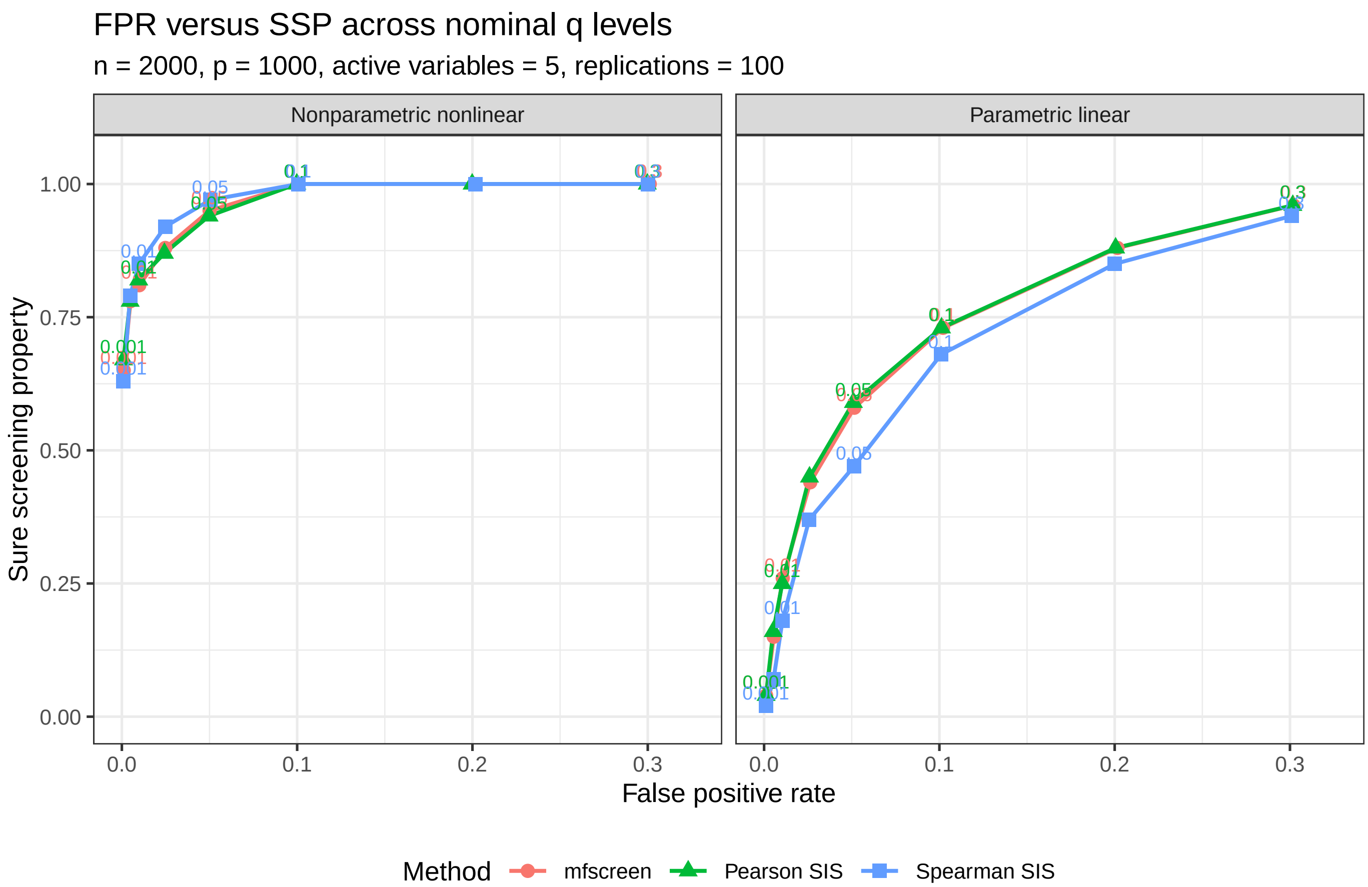
where

$$\sigma(X) = 0.7 + 0.5|X_1| + 0.25X_2^2, \qquad \varepsilon \sim N(0, 0.8^2).$$

This model is nonlinear and heteroskedastic but retains marginal signal in the active variables.

3 Simulation Results

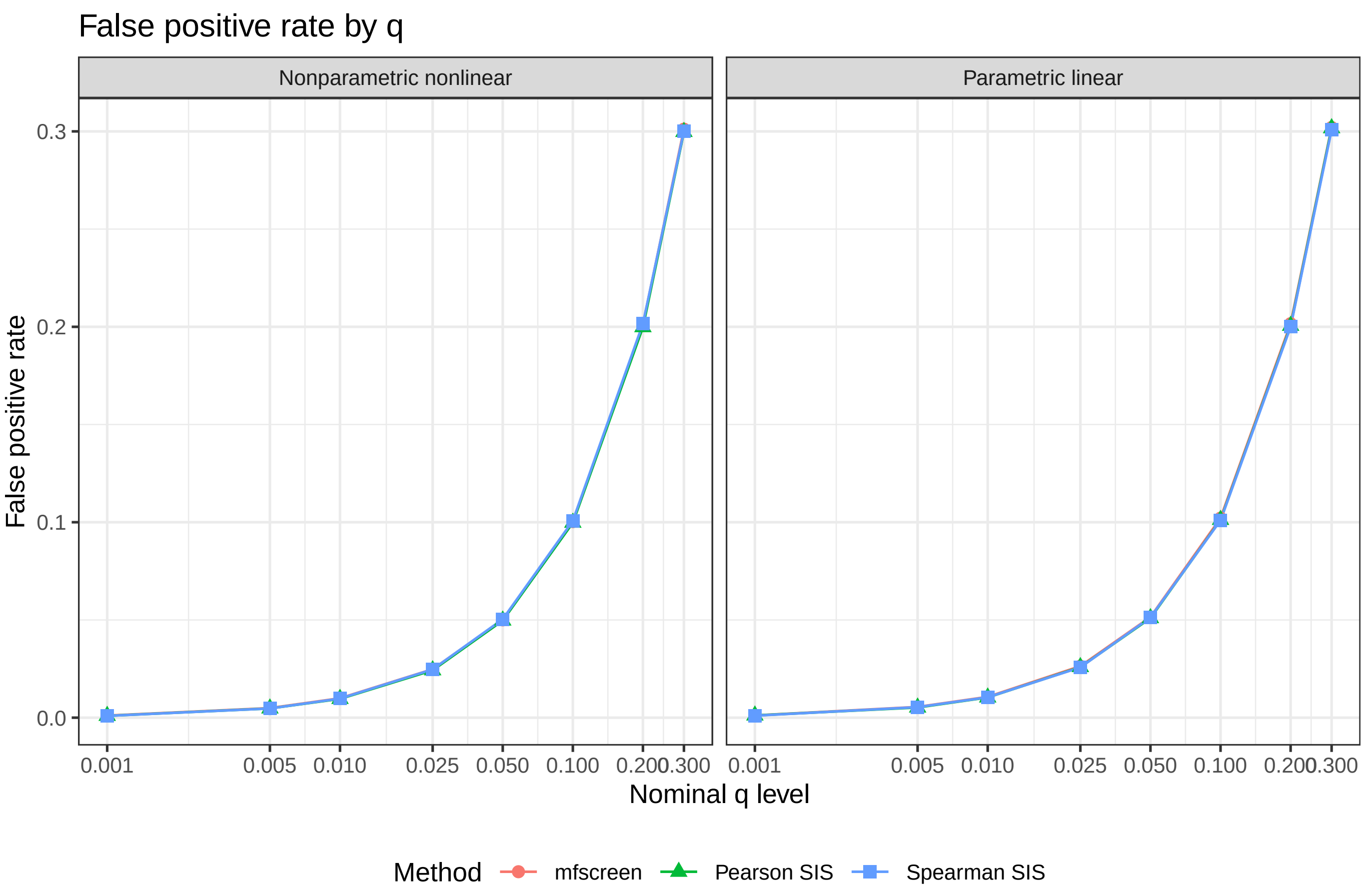
3.1 FPR versus SSP



In the parametric linear model, `mfscreen` and Pearson SIS are very close, as expected: both are driven by marginal linear association. Spearman SIS is less efficient here because ranking discards some linear Gaussian information.

In the nonlinear heteroskedastic model, all three methods retain the active variables once q is moderately large. At smaller q values, the model-free statistic remains competitive with Pearson and Spearman while keeping empirical FPR close to the nominal level. The main advantage of `mfscreen` is that it uses the reciprocal robust statistic rather than the ordinary homoskedastic correlation p-value.

3.2 FPR as a function of q



Across both data-generating mechanisms, the empirical FPR follows the nominal level closely for all three methods. This means that differences in the FPR–SSP plot are primarily differences in sure screening probability rather than failures to control false positives.

This behavior is expected in the baseline experiment. The inactive predictors are independent Gaussian variables, so the two competitors are used in a setting close to the assumptions behind their usual two-sided correlation p-values. In this favorable case, plotting FPR as a function of q mostly checks that the simulation code and thresholding rules are calibrated.

3.3 FPR-control stress test

The preceding baseline does not strongly challenge Pearson or Spearman p-values. To probe FPR control more directly, we also run a global-null stress test with common volatility. In this experiment there are no active variables: the response and all predictors have zero mean association. However, each row shares a random volatility factor

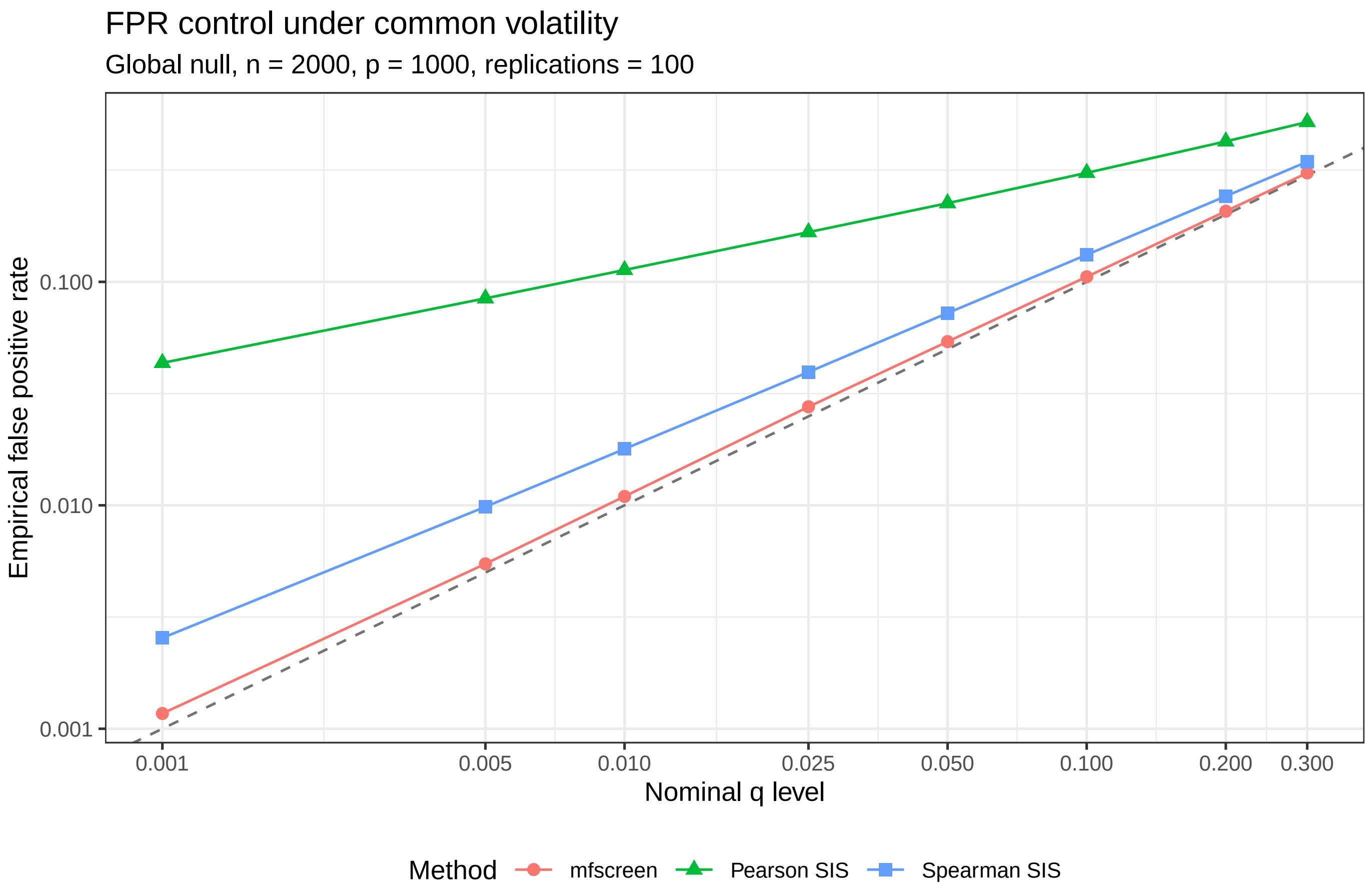
$$A_i = \exp(0.5 Z_i), \qquad Z_i \sim N(0, 1),$$

and

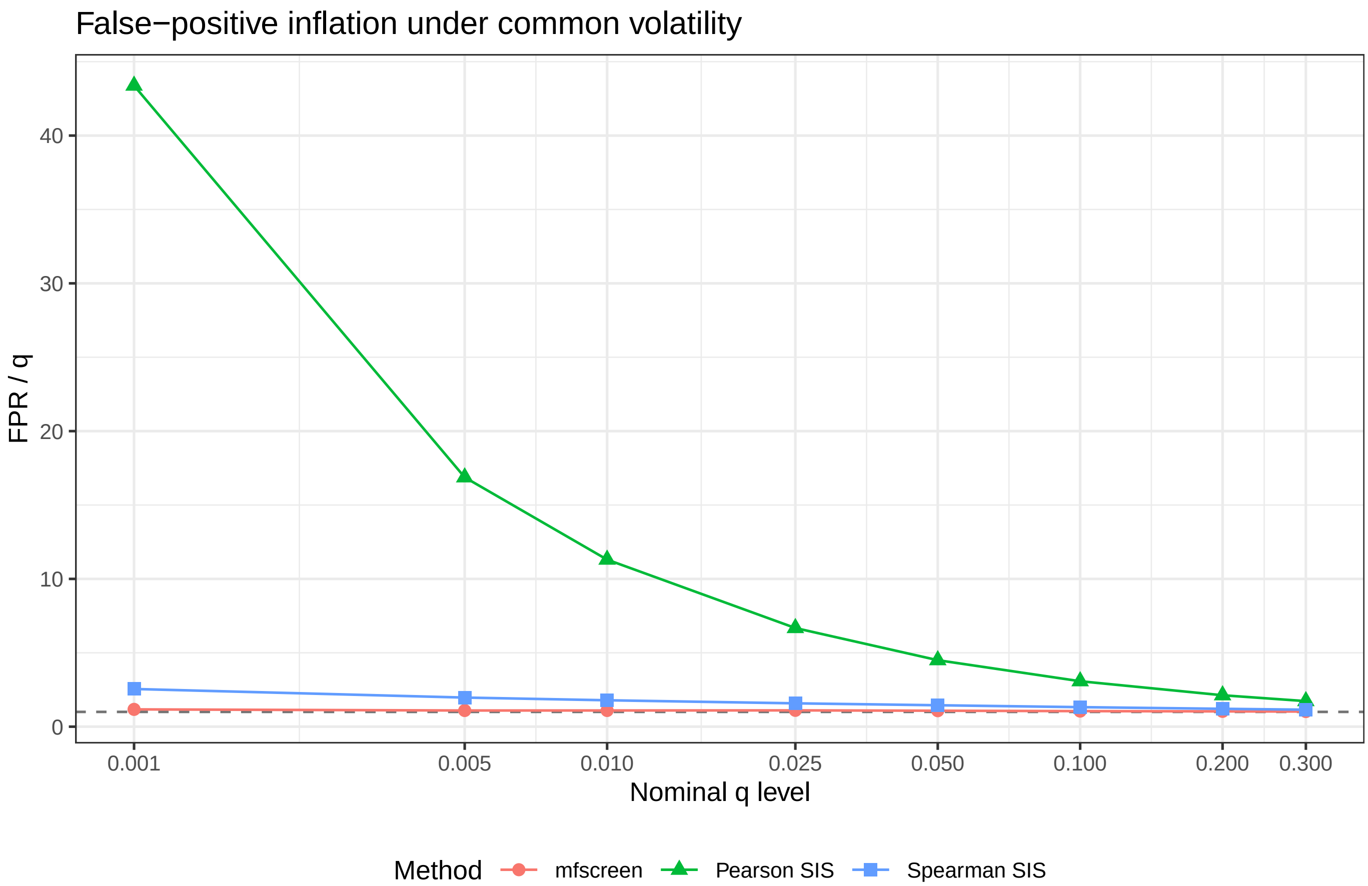
$$X_{ij} = A_i E_{ij}, \qquad Y_i = A_i \varepsilon_i,$$

with independent standard normal E_{ij} and ε_i.

The ordinary Pearson p-value treats observations as homoskedastic and therefore uses the wrong null calibration. Spearman ranks reduce sensitivity to the largest values, but the shared volatility still induces dependence in the joint ranks. The robust reciprocal statistic in `mfscreen` is designed to studentize the marginal slope with a heteroskedasticity-aware denominator, so we expect it to remain much closer to the nominal false positive rate.

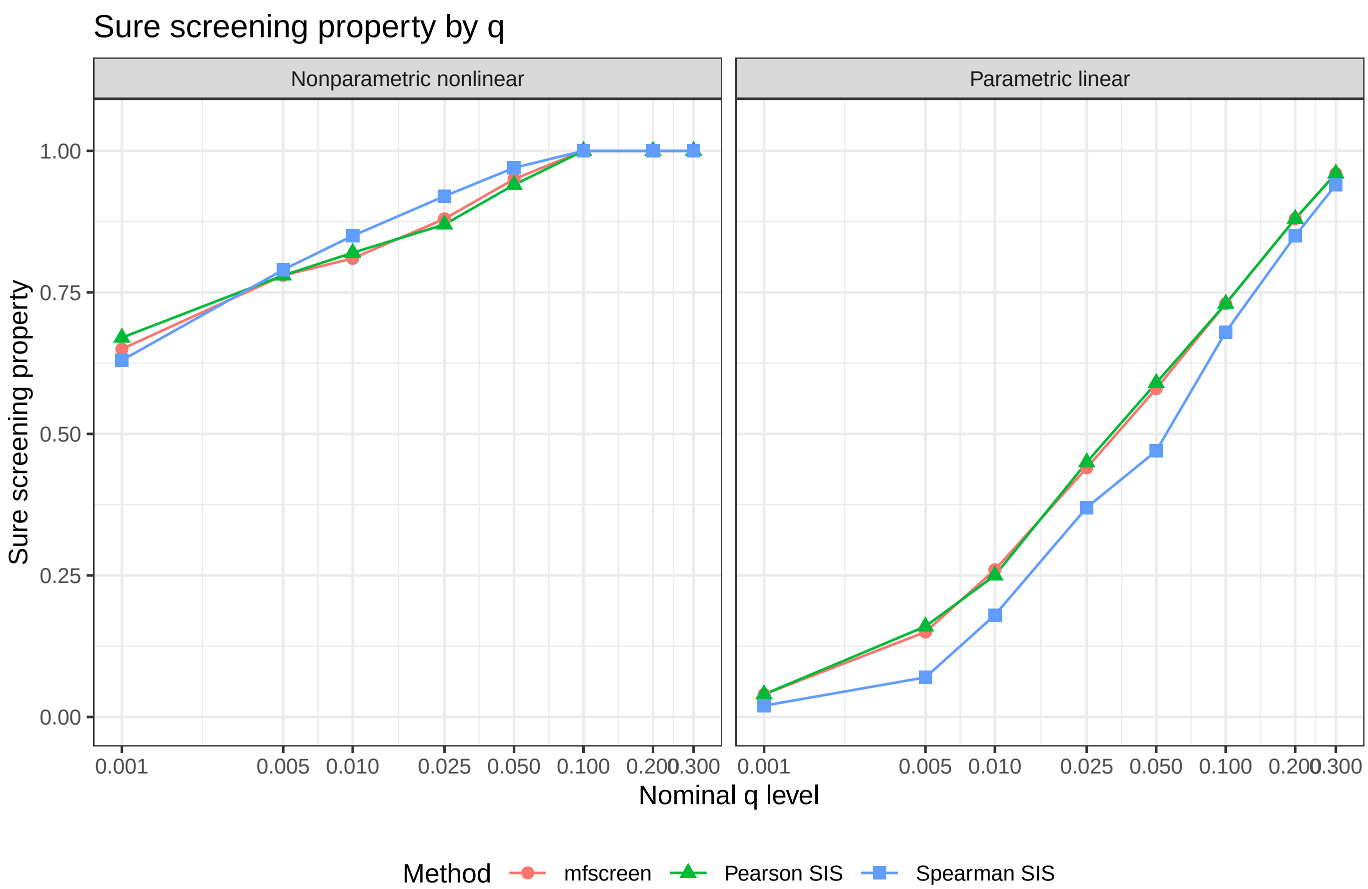


The dashed line is the ideal calibration FPR $= q$. In this stress test, `mfscreen` stays close to the dashed line across the q grid. Pearson SIS is strongly anti-conservative: for example, at $q = 0.05$, its empirical FPR is about 0.225. Spearman SIS is better than Pearson but still inflated: at $q = 0.05$, its empirical FPR is about 0.073. The corresponding `mfscreen` FPR is about 0.054.



The inflation plot makes the calibration gap clearer. Values near one indicate good FPR control. `mfscreen` is close to one, while the two competitors are above one throughout the grid, especially Pearson SIS at small q.

3.4 SSP as a function of q



The SSP curves show the tradeoff directly: increasing q admits more false positives but improves the probability of retaining all active variables. In the linear model, `mfscreen` and Pearson SIS overlap almost exactly. In the nonlinear model, the methods are also close at this sample size, with `mfscreen` competitive across the full q grid and reaching SSP equal to one once $q \geq 0.10$.

4 Numerical Summary

Table 1: Mean FPR and SSP over Monte Carlo replications.

model	q	method	fpr	ssp
Nonparametric nonlinear	0.001	mfscreen	0.0011	0.65
Nonparametric nonlinear	0.001	Pearson SIS	0.0009	0.67
Nonparametric nonlinear	0.001	Spearman SIS	0.0009	0.63
Nonparametric nonlinear	0.005	mfscreen	0.0049	0.78
Nonparametric nonlinear	0.005	Pearson SIS	0.0047	0.78
Nonparametric nonlinear	0.005	Spearman SIS	0.0047	0.79
Nonparametric nonlinear	0.010	mfscreen	0.0099	0.81
Nonparametric nonlinear	0.010	Pearson SIS	0.0096	0.82
Nonparametric nonlinear	0.010	Spearman SIS	0.0098	0.85

model	q	method	fpr	ssp
Nonparametric nonlinear	0.025	mfscreen	0.0248	0.88
Nonparametric nonlinear	0.025	Pearson SIS	0.0243	0.87
Nonparametric nonlinear	0.025	Spearman SIS	0.0247	0.92
Nonparametric nonlinear	0.050	mfscreen	0.0500	0.95
Nonparametric nonlinear	0.050	Pearson SIS	0.0497	0.94
Nonparametric nonlinear	0.050	Spearman SIS	0.0504	0.97
Nonparametric nonlinear	0.100	mfscreen	0.1003	1.00
Nonparametric nonlinear	0.100	Pearson SIS	0.0998	1.00
Nonparametric nonlinear	0.100	Spearman SIS	0.1007	1.00
Nonparametric nonlinear	0.200	mfscreen	0.2011	1.00
Nonparametric nonlinear	0.200	Pearson SIS	0.1998	1.00
Nonparametric nonlinear	0.200	Spearman SIS	0.2016	1.00
Nonparametric nonlinear	0.300	mfscreen	0.3011	1.00
Nonparametric nonlinear	0.300	Pearson SIS	0.2997	1.00
Nonparametric nonlinear	0.300	Spearman SIS	0.3002	1.00
Parametric linear	0.001	mfscreen	0.0011	0.04
Parametric linear	0.001	Pearson SIS	0.0011	0.04
Parametric linear	0.001	Spearman SIS	0.0010	0.02
Parametric linear	0.005	mfscreen	0.0055	0.15
Parametric linear	0.005	Pearson SIS	0.0052	0.16
Parametric linear	0.005	Spearman SIS	0.0054	0.07
Parametric linear	0.010	mfscreen	0.0106	0.26
Parametric linear	0.010	Pearson SIS	0.0104	0.25
Parametric linear	0.010	Spearman SIS	0.0104	0.18
Parametric linear	0.025	mfscreen	0.0263	0.44
Parametric linear	0.025	Pearson SIS	0.0259	0.45
Parametric linear	0.025	Spearman SIS	0.0257	0.37
Parametric linear	0.050	mfscreen	0.0515	0.58
Parametric linear	0.050	Pearson SIS	0.0510	0.59
Parametric linear	0.050	Spearman SIS	0.0513	0.47
Parametric linear	0.100	mfscreen	0.1021	0.73
Parametric linear	0.100	Pearson SIS	0.1012	0.73
Parametric linear	0.100	Spearman SIS	0.1010	0.68
Parametric linear	0.200	mfscreen	0.2016	0.88
Parametric linear	0.200	Pearson SIS	0.2005	0.88
Parametric linear	0.200	Spearman SIS	0.2000	0.85
Parametric linear	0.300	mfscreen	0.3020	0.96
Parametric linear	0.300	Pearson SIS	0.3016	0.96
Parametric linear	0.300	Spearman SIS	0.3009	0.94